

Sentinel (Health & Security Audit App)

Layout Design Map

Assignment 2: The XML Architect

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App Overview

A UI prototype of an Android Security Audit application. It performs a visual Health and Security Audit of apps installed on the user's device.

Layout Requirement Mapping (L1 - L10)

ID	Layout Type	File Name	Function in My App
L1	CoordinatorLayout + AppBarLayout + CollapsingToolbarLayout	activity_main.xml	Main Dashboard. The security score circle and shield logo header collapses into a pinned SENTINEL toolbar when the user scrolls down.
L2	ConstraintLayout (Guideline + Barrier + Chain)	activity_app_detail.xml	App Profile Card. Guideline at 22% separates avatar from text. Barrier prevents text overlap. Chain distributes HIGH RISK / API 21 / CLEARTEXT badges evenly.
L3	LinearLayout (Weighted)	activity_main.xml activity_scan_history.xml	Stats Row showing Total Apps, Threats, and Safe count. Each cell uses layout_weight=1 to ensure equal width on all screen sizes.

L4	RelativeLayout (toRightOf + below + alignParentEnd)	activity_app_detail.xml activity_settings.xml	Permission Detail Rows. layout_toRightOf positions name beside icon, layout_below positions name below icon anchor, layout_alignParentEnd places the status label at the far right.
L5	FrameLayout (layout_gravity = top end)	activity_app_detail.xml	Notification Hub. Bell icon in center. Red circular badge with count 3 is stacked on top using layout_gravity=top end to show unread threat alerts.
L6	GridLayout (columnSpan=2)	activity_main.xml	Quick Actions Grid. RUN FULL SCAN button spans 2 columns using layout_columnSpan=2. Threats and App Audit buttons each occupy one column below.
L7	TableLayout (stretchColumns)	activity_scan_history.xml	Scan History Log table with App Name, Risk Level, and Date columns. stretchColumns=0,1 forces the first two columns to expand and fill the screen width.
L8	HorizontalScrollView	activity_main.xml	Recent Alerts Carousel. Scrolls left-to-right independently of the main vertical NestedScrollView. Each card shows app name, risk level, and permission count.
L9	RadioGroup	activity_settings.xml	Two RadioGroups: Scan Mode (Quick / Deep / Scheduled) and Display Theme (Dark / Light / Follow System). Only one option can be selected at a time in each group.
L10	ConstraintLayout (Flow Helper)	activity_main.xml	Detected Threat Tags Cloud. Flow widget wraps 6 tag chips (#CLEARTEXT, #LOCATION, #CAMERA, #READ_CONTACTS, #MICROPHONE, #STORAGE) to the next line automatically.

Screens Summary

Screen File	Layouts Used	Description
activity_main.xml	L1, L3, L6, L8, L10	Main Dashboard with collapsing header, stats row, quick actions, carousel, tag cloud
activity_app_detail.xml	L2, L4, L5	App Detail with profile card, permission rows, notification badge
activity_scan_history.xml	L7	Scan History with full data table and stretched columns
activity_settings.xml	L9	Settings with scan mode and theme radio groups

XML Drawable Assets

File Name	Type	Used For
bg_card.xml	Shape (Rounded Rect)	Card backgrounds throughout the app
bg_badge.xml	Shape (Oval)	Red circular notification badge on FrameLayout
bg_score_circle.xml	Shape (Oval + Stroke)	Security score circle on Dashboard header
bg_risk_high.xml	Shape (Rounded Rect)	HIGH RISK badge background
bg_surface.xml	Shape + Stroke	Elevated card surface with border
bg_cyan_tag.xml	Shape + Stroke	Threat tag chip backgrounds in Flow tag cloud
bg_cyan_button.xml	Shape (Rounded Rect)	RUN FULL SCAN button background
bg_safe_tag.xml	Shape (Rounded Rect)	Safe status tag background
bg_header_gradient.xml	Gradient Shape	Dashboard header gradient background

bg_glow_ring.xml	Layer-List (3 Ovals)	Triple layered rings around score circle
ic_avatar.xml	Layer-List (Ovals)	App avatar placeholder
ic_shield.xml	Vector Asset	Sentinel shield logo